

No evidence for robot-assembly enhancing adolescents' long-term interactions with social robots

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Abstract

Social robots hold significant promise for classroom education, but high costs and technical complexity hinder widespread adoption. Low-cost, “do-it-yourself” (DIY) robotic toolkits offer a possible solution. The labor of assembly has been shown by prior work to increase positive evaluations of objects (i.e., the IKEA effect) and might increase attachment and long-term engagement. The latter is especially crucial to foster successful, long-term deployment of social robots. However, the assembly process may also reduce socialness perceptions, as those who build robots have “seen behind the curtain”, which can in turn reduce a robot’s educational effectiveness. In this seven-week longitudinal study, we investigated how the process of assembling the humanoid social robot Marty affects students’ ($n = 61$, 12-16 years) perceptions, attachment, and intention to use this robot for future classroom activities. We compared an experimental group that assembled their own robots to a control group that interacted with pre-assembled robots within a digital literacy curriculum. Contrary to the IKEA effect observed in adults and younger children in other studies, our results provide no evidence that assembly changes students’ perceptions of robot socialness, attachment, and intention to use. Exploratory analyses suggest perceived anthropomorphism is the more reliable predictor of students’ perceptions across both conditions. Additionally, initial encounters during weeks 1-4 decreased perceived anthropomorphism and increased safety perceptions among both experimental and control groups. These findings challenge the universality of the IKEA effect in HRI and suggest that for students working with this particular robotic platform, socialness perception is driven more by perceived anthropomorphism than by the labor of assembly. Our work supports the viability of DIY toolkits for school deployment, as we find no evidence that the economic benefits of assembly come at the cost of a robot’s social effectiveness.

1 Introduction and related work

Social robots have shown significant potential to support cognitive and affective learning across diverse educational contexts (Belpaeme et al., 2018). While typically cast as teachers, tutors, or peers, their roles continue to expand to roles including socio-emotional support or conversational instructors (Rohlfing et al., 2022). However, widespread school deployment is hindered by factors including financial burden, technical complexity, user acceptance, and long-term engagement (Barakova et al., 2023; Leisten, Graw, et al., 2026; Leite et al., 2013; Rosén et al., 2022; Woo et al., 2021). One possible solution to these challenges is the use of inexpensive, simple “do-it-yourself” (DIY) social robotic toolkits. Recent platforms such as the Reachy mini (Pollen Robotics & Hugging Face, 2026) or the Blossom robot (Suguitan & Hoffman, 2019) have the ability to perform social behaviors (e.g., movement, speech, facial expressions) at a fraction of the cost (85€-450€) of commonly used preassembled commercial robots like the NAO or Pepper robots (6,000€-17,000€; *Aldebaran Robotics*,

2023; Mahdi et al., 2022). These toolkits often require users to engage in the assembly process, including 3D printing, wiring, motor calibration, and programming. This transition from mere consumers to creators of embodied AI technology holds potential for fostering students' digital literacy and increasing their long-term engagement with these robots, but it also carries risks of breaking the illusion of autonomous social behaviors by these machines, thus reducing the perception of these agents' socialness and thereby reducing their educational effectiveness (Goldman & Poulin-Dubois, 2024). To explore the potential of DIY robots to support students' learning, it is crucial to understand the possible trade-off between the psychological benefits of authorship and the potential loss of socialness attribution.

The process of assembly is well-documented to improve users' evaluation of objects. In 2012, Norton and colleagues (2012) identified the "*IKEA effect*", where manual assembly of purchased IKEA furniture increased consumers' satisfaction with it. The IKEA effect has also been confirmed in a recent meta-analysis (Pelled et al., 2026). Likewise, the "*I designed it myself*" effect suggests that the feeling of authorship enhances perceived value through a sense of ownership and accomplishment (Franke et al., 2010; Sun & Sundar, 2016). In Human–Robot interaction (HRI), numerous studies to date have demonstrated that the sense of ownership and feelings of accomplishment derived from assembling a robot impact users' attachment to it and how people perceive it (Groom et al., 2009; Huang et al., 2013; Sun & Sundar, 2016). For example, Groom and colleagues (2009) show that adults reported lower ratings of maliciousness, higher ratings of friendliness, and stronger team feelings toward a robot they assembled versus a robot assembled by others. In Sun and Sundar's work (2016), adults evaluated both the self-assembled robot and the interaction with it as more positive than a preassembled robot. Assembly has also increased self-efficacy, personal engagement, emotional attachment, and empathy in prior work (Groom et al., 2009; Huang et al., 2013; Tsumura & Yamada, 2025). These findings highlight the positive impact of robot-assembly on users' attachment, which in turn may foster increased intention to use that robot, as well as the long-term engagement needed for successful robot deployment (Zhang & Yu, 2025).

Current social robots are often designed as "black boxes" without clear indications about their functionality, which can affect how users make sense of such technology. According to Dennett's (1971) theory of intentionality, humans adopt different mental stances to predict a system's behavior based on their understanding of such a system. People can take: (i) the physical stance, where they predict behavior based on physical states and natural laws; (ii) the design stance, where they predict behavior based on the system's intended functional design and their knowledge of it; or (iii) the intentional stance, where they treat the system as a social agent with intentional beliefs and desires. Both the physical and design stances require a clear understanding of a system's technical design or functionality (Dennett, 1971). As robots sit along the agent/machine boundary and children lack the technical knowledge and transparency to understand such systems, they are likely to adopt an intentional stance towards social robots (Kahn et al., 2012; UNESCO, 2021; Williams et al., 2019).

Adopting an intentional stance towards social robots may facilitate social attunement and robots' integration into everyday life (Perez-Osorio & Wykowska, 2020). For example, Wiese and colleagues (2012) found that people's social attention is affected by whether or not they take an intentional stance towards robots. Similar findings come from Hechler and colleagues (2025), who showed that believing in human control of artificial systems improved adults' subjective experience in a collaborative task. However, adopting an intentional stance towards technology may not always be beneficial, as it could foster unhealthy emotional attachment or over-trust (Perez-Osorio & Wykowska, 2020) and even result in higher (bodily) wariness in interactions with robots (Broadbent et al., 2011). These risks are particularly problematic for children, who often lack the necessary literacy to interact safely with AI technology (Leisten, Moffat, et al., 2026; Mertala & Fagerlund, 2024; UNESCO, 2021).

The adoption of an intentional stance is not a binary choice, but a dynamic process of socialness attribution (Cross et al., 2016; Hortensius & Cross, 2018). This process is affected by a variety of individual and contextual factors, such as the agent's form and motion, as well as the user's prior beliefs and expectations (Horstmann & Krämer, 2019; Leisten et al., 2025b; Liberman-Pincu et al., 2023). The resulting social perception is multidimensional: we do not judge an agent on a single "social vs. non-social" scale, but across distinct categories, such as experience and agency, warmth and competence, human-nature and human-uniqueness, or other multidimensional approaches (Cross & Ramsey, 2021; Fiske et al., 2007; Gray et al., 2007; Henschel et al., 2020; Leisten et al., 2025b). Because of this multidimensionality, a robot may be perceived as intentional or social on some levels while being understood as a functional tool on others. Consequently, robot-building may support the adoption of a design stance by learning about the system's underlying functions (Hitron et al., 2019; van Straten, Peter, et al., 2020), while the time and effort spent on assembly may create feelings of ownership and attachment (Sun & Sundar, 2016). By allowing children to understand the machine, while still attaching to the social agent, social DIY robots may be particularly well-suited for responsible and beneficial school deployment.

Despite the potential tension between psychological ownership and socialness perception, very few studies have investigated the effect of assembly on children's perceptions of robots. Williams and colleagues (2019) collected pre-schoolers' (4-6 years) perceptions of a robot's theory of mind, agency, intelligence, and socialness before (pre-test) and after (post-test) building a PopBots robot (consisting of LEGO blocks, a phone, and a tablet) and participating in AI learning activities. From pre- to post-test, children's perception of a robot's intelligence and socialness increased. These effects were moderated by children's AI knowledge: children with higher AI knowledge were more likely to perceive robots as intelligent and social. Su and Yang (2024) extended these results by assessing how an AI literacy curriculum may affect pre-schoolers' perceptions of robots using the same measures as Williams and colleagues (2019). The authors found that children perceived a robot as more intelligent and social after learning about AI. This research sets the stage for investigations into the influence of robot-building on older children and adolescents' perceptions of robots' socialness. Adolescents are an

especially relevant stakeholder group in this context, as they are at particular risk from unsafe AI interactions, due to their high usage of the technology but insufficient AI education and risk awareness (Sarikakis & Chatziefrimidou, 2025; UNESCO, 2021). To evaluate the potential of robot-assembly kits to overcome robot deployment challenges (i.e., costs, technical complexity, long-term engagement), it is important to assess the influence of robot-assembly on middle-school children's and adolescents' perceptions, attachment, and intention to use this robot.

1.1 The present study

In this study, we sought to investigate how building a social robot across multiple classroom lessons influences older children's and adolescents' perceptions of, attachment to, and intention to use a social robot. In one of the first long-term educational HRI studies with adolescents, we collected survey data at three time points during a seven-week educational curriculum for students aged 12-16 years. To isolate the influence of the robot-building experience, we employed an active control group design with a *build group*, who built robots, and a *control group*, who interacted with already assembled robots. We assessed students' perceptions of the robots they built (build group) or interacted with (control group), as well as their perceptions of two other robots they did not build/interact with. Additionally, we assessed students' attachment and intention to use the robot they built/interacted with. We aimed to address the following research questions:

- **RQ1:** What is the effect of robot-building on students' social perception of robots?
- **RQ2:** What is the effect of robot-building on students' attachment and intention to use the robot?

In line with the above-reviewed work, we expected students in the build group to perceive the robot as more competent (**H1**), warmer (**H2**), safer (**H3**), and less discomfoting (**H4**), and more anthropomorphic (**H5**), over time than in the control group. We also expected students in the build group to feel more attached to (**H6**) and willing to use (**H7**) the robot over time than students in the control group. We additionally preregistered exploratory analyses of interactions with students' prior digital literacy knowledge and attachment to the robot.

2 Methods and materials

2.1 Participants

We recruited a convenience sample of two participating classes from a rural German comprehensive school (i.e., including students from the three German academic levels Hauptschule, Realschule, and Gymnasium), resulting in a sample of 61 students between 12-16 years (15 females, 42 males, 4 other/NA, $M_{age} = 13.40$, $SD = 0.75$). Participants were native German speakers. Informed parental consent was collected, and students aged 14 and older provided additional written assent. Additional

consent/assent was obtained to take and publish photos. The ETH Zurich's ethical committee approved all experimental procedures before data collection began (25 ETHICS-247).

2.2 Design and procedure

We employed a longitudinal, between-subjects, active-control design (3x2 with the factors condition (experimental vs. control) and time point (T1, T4, T7)), conducted over a seven-week educational curriculum. The experiment was conducted during students' regular Natural Sciences and Informatics class and split into two phases: in phase 1 (weeks 1-4), students in the build group ($n = 31$) built their own social robot, *Marty* (Robotical, 2025), in randomly assigned teams of 3-4. Students stayed within their assigned teams throughout the entire curriculum. Students in the control group ($n = 30$) interacted with pre-assembled Marty robots, also in teams of 3-4. In phase 2 (weeks 5-7), both groups used their respective robots to engage in identical learning interactions. We selected robot interaction as the control condition to ensure interaction time was consistent across conditions. This allowed us to compare robot assembly to typical robot deployment, namely, to pre-assembled tools that students can readily interact with.

The seven-week curriculum was co-developed with teachers, based on prominent AI literacy curricula (ADAPT center, 2026; *AI Pedagogy Project*, 2026; *AI4GA*, 2026; *AI4K12*, 2026; *aiEDU*, 2026; *Day of AI*, 2026; Common sense media, 2026; Computingeducation, 2025; Duke Center for Teaching and Learning, 2026; MIT Media Lab, 2026; MIT RAISE, 2026; OECD, 2025; Roteco, 2026; Toddle, 2026), and covered social robotics, Artificial Intelligence (AI), sensors and microcontrollers, Large Language Models (LLMs), and programming (see **Figure 1** for content). The curriculum development process is described in more detail in (Leisten, Muss, et al., in prep.), and the curriculum is available on GitHub in English and German (Muss & Leisten, 2026). Each weekly session included theoretical content and practical activities and lasted 140 minutes (see **Figure 1** for timings). The total experimental time in the classroom was 16 hours and 33 minutes. Participants were divided into two parallel tracks of 16 students led by two separate teachers, due to the school's classroom set-up with two adjacent rooms. We first conducted two experimental groups in parallel (Nov – Dec 2025), followed by two parallel control groups (Jan – Mar 2026).

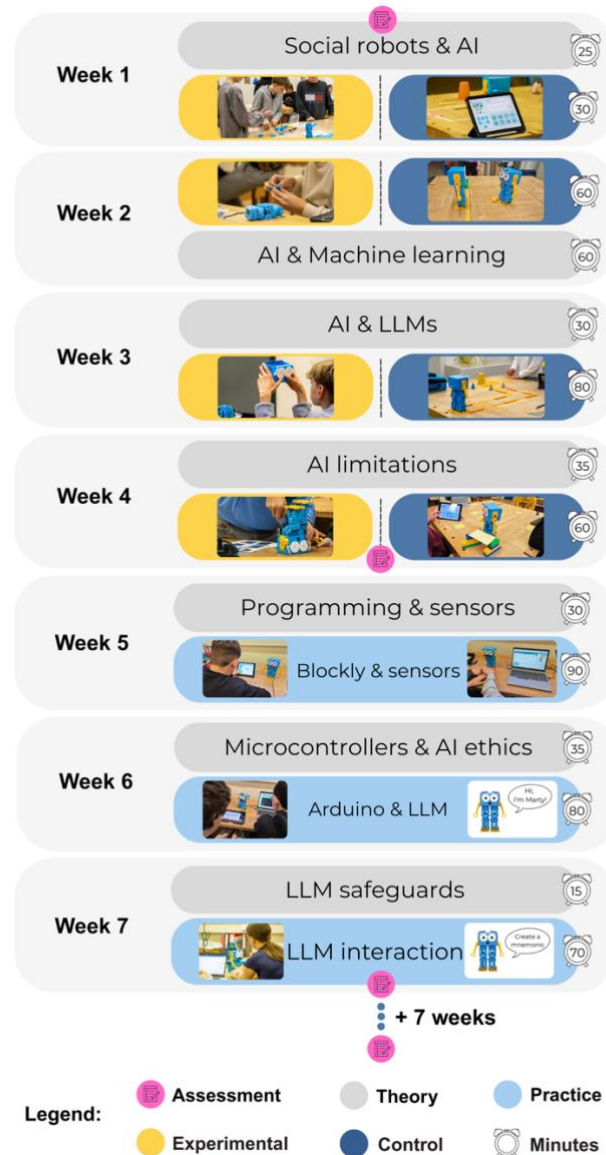


Figure 1. Schematic overview of the seven-week curriculum.

During the practical activities in weeks 1-4, students in the building group assembled their Marty robots following a build guide (see Leisten et al., 2025a). Students started with the legs (weeks 1 and 2), continued with the arms (week 3), and finished the assembly in week 4. The total build duration was approximately four hours. Students in the control group interacted with a preassembled Marty instead, familiarizing themselves with the App and control mechanisms (week 1), playing a soccer tournament (week 2), building and completing an obstacle course (week 3), and playing an escape game (week 4; see **Figure 2** for pictures). In these activities, students navigated their Marty robots using the Marty controller without requiring any programming. We designed these activities specifically to be engaging but not to provide any relevant learning content about Marty or AI, to not give this group a theoretical advantage over the building group, while keeping the interaction time with Marty consistent.



Figure 2. Deployment pictures. A-D) Students building the Marty robot. E) Theoretical content delivered by the teacher. F) Student navigating Marty in a penalty shoot-out. G) Marty navigating through an obstacle course. H) A teacher helping a student with Blockly programming.

Students completed pre-, mid-, and post-test surveys (~15 minutes) in Week 1 (T1), Week 4 (T4), and Week 7 (T7) via Qualtrics (2024) surveys on iPads. In the surveys, students first watched 4-5-second-long videos of Marty, NAO, and Blossom, and rated each robot on seven social attributes using the robotic social attributes scale (RoSAS-SF; Carpinella et al., 2017; Neuenswander et al., 2025). We selected NAO and Blossom, in addition to Marty, to provide examples of a widely used, complex robot (NAO) and a simpler DIY robot (Blossom). The video stimuli were obtained from YouTube and muted to avoid confounding effects (for edited videos, see Leisten, Caruana, et al., 2024; Robotical, 2019; YouTube, 2024). The order of robots and attributes was counterbalanced. Next, students were presented with a picture of Marty and asked to answer the specific object anthropomorphism scale (SOAS; David et al., 2025), the product-consumer attachment scale (Schifferstein & Zwartkruis-Pelgrim, 2008), and the intentional acceptance scale (De Jong et al., 2020). Scale and item order were counterbalanced. Lastly, students were asked about their subjective AI literacy knowledge and their objective digital literacy (including technical, programming, and AI literacy), and additionally completed short feedback surveys (~10 minutes) at the end of each session, all of which we report elsewhere (Leisten, Jordan, et al., in prep.). Students were naïve about the study objectives and were debriefed after both groups completed the study. Teachers received a 100€ voucher towards classroom materials for their classes' participation. All our methods, research questions, and analyses were preregistered, and the data, code, and materials are openly available on the OSF (Leisten et al., 2025a).

2.3 Marty robot

Marty is a medium-sized (~25 cm), desktop humanoid robot, designed to support students' STEAM (Science, Technology, Engineering, Arts, Mathematics) education at a price point of ~400€ (Robotical, 2025). Marty is equipped with two LED eyes, eleven motors (eyes, arms, hips, knees, thighs), and five

sensors (acceleration, tilt, position, color, infrared), and is usually sold preassembled. However, Marty can also be assembled using a build guide (Figure 23; see Leisten et al., 2025a), which is estimated to take around 2-3 hours. The building process consists of connecting motors, cables, and sensors to on-board computers, screwing together plastic parts, calibrating arms and eyes, and mounting the legs, arms, and eyes to the head. Once assembled, Marty can walk, turn, dance, celebrate, speak, play music, eye-wiggle, and ball-kick through screen-free, block-based, or text-based coding (Python or C++ with Arduino) using an App or web-based version.

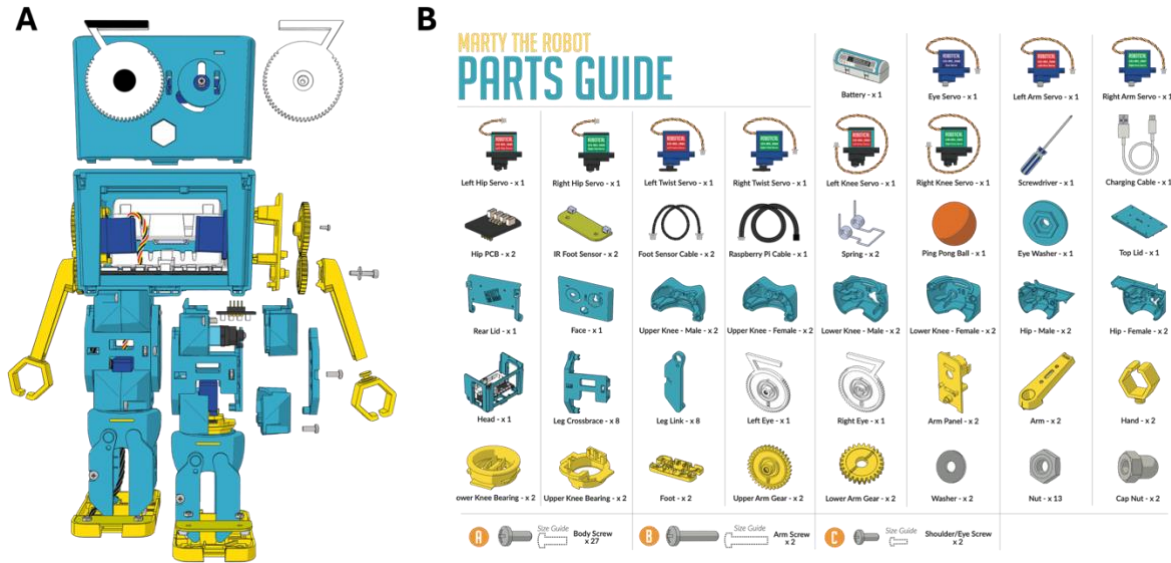


Figure 3. A) Schematic build guide of the Marty robot (copyright belongs to Robotical). B) Overview of parts guide (copyright belongs to Robotical).

2.4 Measures

Robotic social attributes scale (RoSAS-SF). We measured students' social perception of Marty using the validated short version of the RoSAS scale (RoSAS-SF; Carpinella et al., 2017; Kühne et al., 2024; Neuenswander et al., 2025). The RoSAS-SF is a 6-item scale for assessing a robot's perceived warmth (compassionate, social), competence (competent, reliable), and discomfort (scary, awkward). We additionally added an item for safety, as robot safety is a concept rarely addressed in educational HRI (Woo et al., 2021) and is especially relevant in interactions with students. Students were asked to rate the robot on all seven attributes using a 7-point Likert scale (e.g., "*Marty is compassionate*"; 1 = *not at all*, 7 = *very much so*). The reliability of the RoSAS-SF has been established (warmth: $\alpha = .85$; competence: $\alpha = .92$; discomfort: $\alpha = .68$; Neuenswander et al., 2025).

Specific object anthropomorphism (SOAS). In the absence of a short, text-based measure of students' intentional stance towards robots, we used the newly developed and validated SOAS scale (David et al., 2025) to measure students' perceived anthropomorphism of Marty as an indicator of their intentional

stance towards it. The SOAS is a six-item scale, developed to assess the extent to which a specific object (in this case, Marty) possesses anthropomorphic qualities (e.g., “*I feel that this object has likes and dislikes*”). We added an additional item asking to what extent Marty has a mind with intentions. Students rated the seven items on a 3-point Likert scale (0 = *no*, 1 = *maybe*, 2 = *yes*). Test-retest and internal reliability of the SOAS have been established ($\omega = .94$, ICC = .78; David et al., 2025).

Consumer-product attachment scale. We assessed students’ attachment to Marty with the consumer-product attachment scale developed by Schifferstein and Zwarthuis-Pelgrim (2008). The scale was originally developed to assess people’s attachment to products they own (e.g., a lamp, a clock), and has also been deployed to HRI (Kim et al., 2013). We opted to use this measure instead of other attachment measures that focus more on human-like relationship formation (e.g., van Straten, Kühne, et al., 2020), as we agree with recent concerns about unhealthy emotional attachment to robots (e.g., Law et al., 2022; Rabb et al., 2022), and believe such attachment should be treated similarly to humans’ attachment to other objects rather than other humans. The attachment construct of the scale comprises five items (e.g., “*I feel emotionally connected to Marty*”), measured on a 5-point Likert scale (1 = *strongly disagree*, 5 = *strongly agree*). The reliability of the scale is excellent ($\alpha = .92$; Kim et al., 2013).

Intentional acceptance scale. Lastly, we measured students’ intention to use Marty with the intentional acceptance scale developed by De Jong and colleagues (2020). Students answered the 4-item scale on a 5-point Likert scale, which we adapted to match the aforementioned attachment scale (1 = *strongly disagree*, 5 *strongly agree*). The scale’s reliability was found to be acceptable ($\alpha = .72$; De Jong et al., 2020).

2.5 Data analysis

Prior to hypothesis testing, composite scores were aggregated for each outcome variable by averaging (competence, warmth, discomfort, safety, attachment, and intention to use) or summing (anthropomorphism) the respective questionnaire items. To examine changes over time and differences between conditions, we used linear mixed-effects models (LMER; lme4, Bates et al., 2015). We accounted for the nested structure of the data by including random effects for participants nested within triads. We preregistered models with the main effects of condition and timepoint, as well as their interaction. To prevent overfitting and reduce the risk of false positives, we incrementally added effects and ran model comparisons using likelihood ratio tests. Where a significant increase in explained variance was observed, we computed post hoc contrasts using estimated marginal means (Lenth, 2024) with false discovery rate (FDR) correction. We additionally preregistered exploratory analyses in which we extended the models to include prior knowledge, perceived anthropomorphism, and attachment as covariates. These were evaluated using the same model comparison approach described above. We additionally preregistered exploratory analyses on the generalizability of effects to other robots (NAO

and Blossom), but decided not to run them. All analyses output can be found in our OSF project (Leisten et al., 2025a).

3 Results

3.1 RQ1: Perception

3.1.1 Competence (H1)

Likelihood ratio tests showed no evidence that condition ($\chi^2 = 1.53, p = .22$), timepoint ($\chi^2 = 0.37, p = .83$), or their interaction ($\chi^2 = 0.39, p = .82$) predicted perceived competence (**Figure 4A**). This was confirmed by our preregistered model, which did not improve model fit.

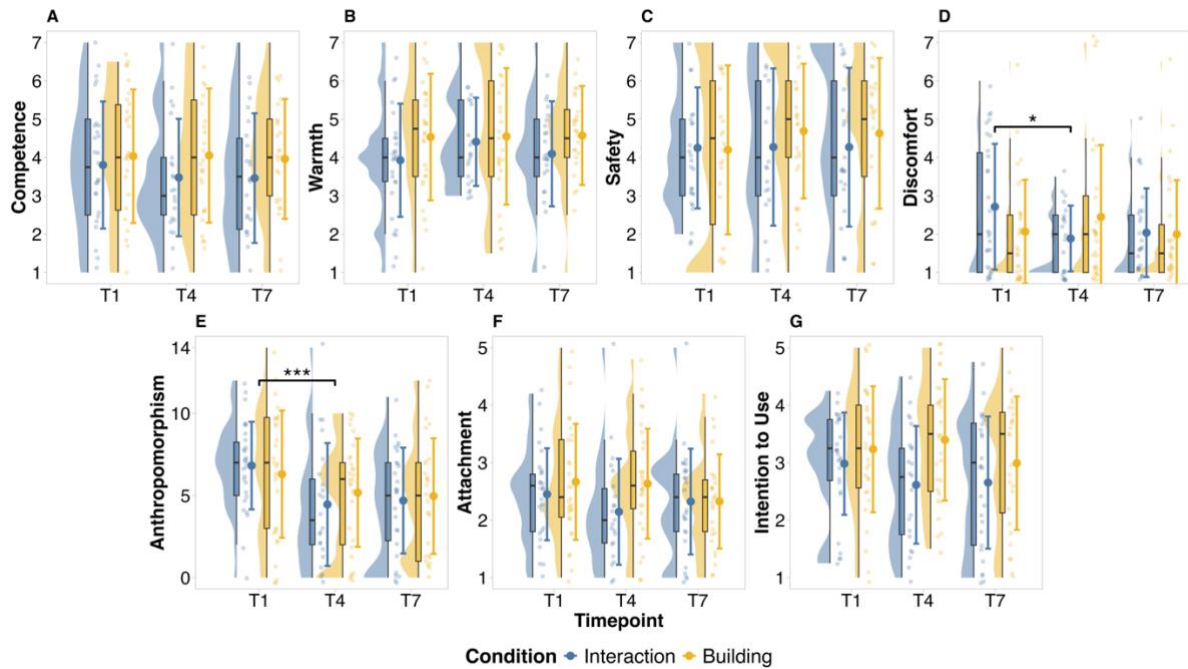


Figure 4. Means, standard deviations, and raw data points of study variables (RQ1 & RQ2), by group and condition. A) Non-significant effects on perceived competence. B) Non-significant effects on perceived warmth. C) Non-significant effects on perceived safety. D) Significant interaction effect of time and condition on perceived discomfort. E) Significant decrease of anthropomorphism across time. F) Non-significant effects on perceived attachment. G) Non-significant effects on intention to use.

Preregistered exploratory analyses. Students' prior digital literacy knowledge was not associated with perceived competence ($\chi^2 = 0.11, p = .75$). However, perceived anthropomorphism significantly predicted variance in competence ($\chi^2 = 23.56, p < .001$), indicating that students who perceived the robot as more human-like regarded it to be more competent ($\beta = 0.19, SE = 0.04, p < .001$), irrespectively of condition or timepoint (**Figure 5A**).

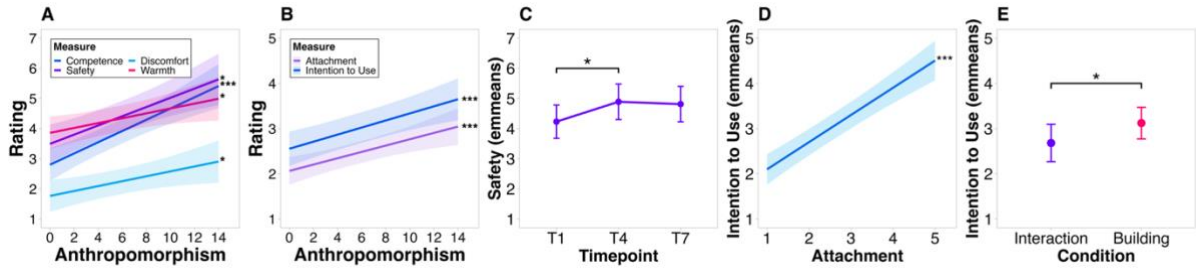


Figure 5. Model-based estimated marginal means of study variables. A-F) Significant increase in outcome variables (y-axes) with increasing anthropomorphism. G) Main effect of timepoint of safety, when controlling for anthropomorphism. H) Significant increase in intention to use with increasing attachment. I) Main effect of condition on intention to use, when controlling for attachment.

3.1.2 Warmth (H2)

Similar to competence, likelihood ratio tests indicated students' perceived warmth was not predicted by condition ($\chi^2 = 2.78, p = .10$), timepoint ($\chi^2 = 1.11, p = .58$), or their interaction ($\chi^2 = 1.60, p = .45$; **Figure 4B**). This was confirmed by our preregistered model, which did not improve model fit.

Preregistered exploratory analyses. Prior knowledge did not explain variance in students' warmth perception ($\chi^2 = 0, p = .98$). Anthropomorphism positively predicted perceived warmth ($\chi^2 = 5.02, p = .03$), with students perceiving the robot as warmer when they attributed more human-like qualities to it ($\beta = 0.08, SE = 0.04, p = .02$), independent of time and group (**Figure 5A**).

3.1.3 Safety (H3)

In lieu of previous perception measures, perceived safety was not affected by condition ($\chi^2 = 0.71, p = .38$), timepoint ($\chi^2 = 2.06, p = .36$), or their interaction ($\chi^2 = 1.14, p = .56$; **Figure 4C**). This was confirmed by our preregistered model, which did not improve model fit.

Preregistered exploratory analyses. Students' prior digital literacy knowledge did not affect the perceived safety of the robot ($\chi^2 = 0.17, p = .68$). However, adding anthropomorphism and timepoint significantly improved model fit ($\chi^2 = 7.91, p = .02$), with both variables positively predicting perceived safety (**Figure 5A**). Pairwise contrasts showed an increase in perceived safety between T1 and T4 ($\beta = 0.66, SE = 0.26, p = .02$), but not between T4 and T7 ($\beta = -0.08, SE = 0.25, p = .76$; **Figure 4C**).

3.1.4 Discomfort (H4)

Likelihood ratio tests showed no main effect of condition ($\chi^2 = 0.03, p = .87$) or time point ($\chi^2 = 3.56, p = .17$) on perceived discomfort. However, the interaction between condition and timepoint significantly improved the model ($\chi^2 = 7.35, p = .03$). Post-hoc contrasts calculated from our

preregistered model indicated a significant difference in trajectories of discomfort in the control and build groups between T1 and T4 ($\beta = -1.14$, $SE = 0.43$, $p = .049$). Specifically, discomfort significantly decreased between T1 to T4 in the control group ($\beta = -0.79$, $SE = 0.33$, $p = .03$), but not in the build group ($\beta = 0.35$, $SE = 0.29$, $p = .23$; **Figure 4D**). However, because both groups did not differ significantly in perceived discomfort at T1 or T4, this interaction should be interpreted with caution.

Preregistered exploratory analyses. Prior knowledge did not explain variance in students' discomfort perception ($\chi^2 = 0.01$, $p = .93$). However, perceived anthropomorphism significantly improved model fit relative to the baseline model ($\chi^2 = 5.66$, $p = .02$). Higher perceived anthropomorphism was associated with increased perceived discomfort ($\beta = 0.08$, $SE = 0.03$, $p = .02$), irrespective of group or time (**Figure 5A**).

3.1.5 Anthropomorphism (H5)

Distinct from our other perception measures, we found evidence for a significant effect of time on perceived anthropomorphism ($\chi^2 = 19.33$, $p < .001$), but no effect of condition ($\chi^2 = 0.05$, $p = .83$), or their interaction ($\chi^2 = 1.55$, $p = .46$). Post-hoc contrasts indicated a decrease in anthropomorphism between T1 and T4 ($\beta = -1.68$, $SE = 0.43$, $p < .001$), but not between T4 and T7 ($\beta = 0.04$, $SE = 0.45$, $p = .93$; **Figure 4E**).

Preregistered exploratory analyses. Preregistered exploratory analyses indicated that prior knowledge did not explain variance in perceived anthropomorphism ($\chi^2 = 1.08$, $p = .30$).

3.2 RQ2: Attachment and intention to use

3.2.1 Attachment (H6)

Students' attachment to the robot was not significantly affected by condition ($\chi^2 = 1.21$, $p = .27$), time ($\chi^2 = 4.31$, $p = .12$), or their interaction ($\chi^2 = 0.82$, $p = .66$; **Figure 4F**). This was confirmed by our preregistered model, which did not increase model fit.

Exploratory analyses. Students' prior digital literacy knowledge did not significantly predict their attachment to the robot ($\chi^2 < 0.001$, $p = .98$). Perceived anthropomorphism significantly improved model fit relative to the baseline model ($\chi^2 = 11.53$, $p < .001$), with higher perceived anthropomorphism being associated with greater attachment ($\beta = 0.07$, $SE = 0.02$, $p < .001$), irrespective of group or time (**Figure 5B**).

3.2.2 Intention to use (H7)

Students' intention to use was not significantly predicted by condition ($\chi^2 = 3.48, p = .06$), timepoint ($\chi^2 = 5.70, p = .06$), or their interaction ($\chi^2 = 2.91, p = .23$; **Figure 4G**).

Preregistered exploratory analyses. Attachment and condition significantly improved model fit ($\chi^2 = 4.46, p = .04$), with attachment positively predicting students' intention to use the robot ($\beta = 0.60, SE = 0.07, p < .001$), irrespective of condition or timepoint (**Figure 5D**). Students in the building group showed a higher intention to use the robot than students in the control group ($\beta = 0.44, SE = 0.19, p = .02$; **Figure 5E**). However, since this main effect also includes students' pre-test scores, for which we would not expect group differences, it should be interpreted with caution.

Exploratory analyses. Anthropomorphism significantly predicted intention to use ($\chi^2 = 14.23, p < .001$), with higher ratings associated with greater intention to use the robot, irrespective of condition or time point (**Figure 5B**).

4 Discussion

In this seven-week, classroom-based HRI study, we investigated how building a social robot affects students' perceptions, attachment, and intention to use the Marty robot. By comparing a building group with an interaction group within a digital literacy curriculum, we sought to determine whether the labor of manual assembly produces an IKEA effect in a social robotics context. Our primary finding is the absence of this effect: we observed no significant differences between manual assembly and pre-assembled conditions in students' ratings of perceived competence, warmth, safety (**RQ1**), attachment, or intention to use the Marty robot (**RQ2**). Instead, our exploratory analyses suggest that anthropomorphism is a more robust driver of these perceptions than the assembly process itself. Furthermore, we observed that initial encounters with the robot during the building/interaction phase led to a recalibration of perceptions: students rated the robot as less anthropomorphic but safer. We discuss the implications of these results below.

4.1 Re-evaluating the IKEA effect in Adolescent HRI (RQ1 & RQ2)

Contrary to our preregistered hypotheses (H1-H7), our results provide no evidence that assembling a robot increases students' perceptions of socialness, attachment, or intention to use the robot. These results contrast with previous findings of the IKEA effect in HRI, where assembly has been consistently reported to produce more favorable evaluations of the built robot (Groom et al., 2009; Huang et al., 2013; Sun & Sundar, 2016; Williams et al., 2019). We propose that this discrepancy is not a failure of

the DIY model, but rather a reflection of the specific intersection of robot morphology, task/context, and user development.

The robot. Unlike studies using LEGO-based platforms (Groom et al., 2009; Tsumura & Yamada, 2025; Williams et al., 2019), we used a humanoid robot. Prior research indicates that the appearance of robots significantly affects social expectations (Liberman-Pincu et al., 2023; Nijssen et al., 2021). Marty's humanoid appearance may have triggered a ceiling effect, leaving little room for assembly to further enhance perceived socialness. Additionally, robots differ from furniture in that they have not only a hardware component but also a software component. Only letting children assemble the hardware, without involving them in “building” the software, might have only been half a building activity. Other work involved children in the training of a robot's AI algorithm (Wang et al., 2026) or in building the underlying program (Sun & Sundar, 2016), which offer promising pathways to combine hardware and software building for future work.

The sample's age. Second, the age of our sample likely played an equally important role. Most existing HRI assembly studies focus on adults (Groom et al., 2009; Huang et al., 2013; Sun & Sundar, 2016; Tsumura & Yamada, 2025) or preschool-aged children. Williams and colleagues (2019) present one of the only studies investigating the impact of robot assembly on *children's* perceptions of robots. However, children in William's sample were significantly younger (4-6-year-olds) than in our sample. Age is an important driver behind differences in robot perception and interaction (e.g., Leisten, Heyselaar, et al., 2024; Manzi et al., 2020; Morillo-Mendez et al., 2024). For example, Leisten and colleagues (2024) found quadratic relationships between children's age (5-12 years) and their perceived closeness and trust with a robot. Similarly, Manzi and colleagues (2020) highlight a decrease in perceived anthropomorphism between ages 5 and 9. Work on the developmental emergence of the IKEA effect supports this explanation, as Marsh and colleagues (2018) report differences in the IKEA effect between children and adults. Specifically, in their series of studies of 5-6-year-olds, the IKEA effect was not fully explained by feelings of ownership, unlike in adult samples (Groom et al., 2009; Huang et al., 2013; Sun & Sundar, 2016). Additionally, while adults are negatively affected by the process costs of building, this mediation does not appear to hold for children (Marsh et al., 2018). For our 12–16-year-old participants, this lack of effect may reflect a developmental shift toward more utilitarian or task-oriented evaluations of technology. Consequently, the labor of assembly may have been perceived through the lens of a common classroom task in their technical class, rather than as a self-expressive act of creation that triggers an emotional bond. This may suggest that the IKEA effect diminishes during adolescence, as teenagers' focus on functional outcomes overrides the ownership-through-effort mechanisms observed in younger children.

The comparison/task. Lastly, unlike prior work that compared building to another building task (e.g., humanoid robot vs. car robot; Groom et al., 2009), used software building/programming rather than

hardware-building (Sun & Sundar, 2016), or used pre-/post-test designs with no comparison group (Huang et al., 2013; Tsumura & Yamada, 2025; Williams et al., 2019), we compared robot-building to an active, gamified robot-operation task. It is possible that the high level of engagement in the control group's interactive tasks provided a similar sense of accomplishment to the building task, thereby neutralizing the IKEA effect of assembly. Furthermore, our intervention focused on instruction-based, group assembly rather than individual personalization. Personalization has been linked to psychological ownership and self-extension (Voges et al., 2026), and its relative absence in our building task may explain why we did not find the expected effects. Future work should aim to disentangle these effects by assessing how individual personalization within the building process might shift students' perceptions.

4.2 Shifts in perceived discomfort and intention to use

Beyond broader perceptions of socialness, our results indicated that students within the control group reported less perceived discomfort with Marty from pre-test to mid-test. This shift was not observed in the experimental group. We tentatively propose that this could indicate a positive effect of robot play in reducing discomfort. However, we must note that perceived discomfort did not differ significantly between the control and experimental groups at either the pre-test or the mid-test. This suggests that while the interaction group's experience with a pre-assembled robot may have reduced discomfort, we did not find evidence that it made the robot significantly more comfortable than in the build group. Future research should examine whether assembly reduces or amplifies user discomfort.

Additionally, our results indicated a higher intention to use among the experimental group than the control group across all time points. As we did not observe evidence of variation across time points, and the effect also includes students' pre-test scores, we cannot confidently interpret this as a higher intention to use a robot when building it. Future research is needed to reassess the effect of robot assembly on students' intention to use such a robot.

4.3 First encounters with a robot drive changes in perceived anthropomorphism and safety

Our study offers longitudinal insights into how students' perceptions of a social robot evolve during a school-based intervention. We observed a decrease in anthropomorphism within the first four weeks, regardless of whether the students built the robot. This change may reflect a shift in students' technical understanding of the robots' capabilities. Prior work suggests that experiences with robots can increase teenagers' positive evaluations of a robot's helpfulness and perceived comfort (Björling et al., 2019), as well as alter the perceived relevance of a robot's capabilities (Sciutti et al., 2014). Van Straten and colleagues extended this work to assess how transparency about a robot's mechanical nature affects perception. Across multiple studies, the authors found decreased anthropomorphism (van Straten et al.,

2022; van Straten, Peter, et al., 2020), but no effects on perceptions of socialness or feelings of companionship towards this robot (Turkle et al., 2006; van Straten et al., 2022). These findings suggest that informing students about a robot's mechanical nature reduces anthropomorphism without diminishing perceptions of socialness. Our findings confirm and extend this knowledge by suggesting that the same mechanisms may also apply to samples of older children and adolescents. Our findings additionally suggest that exposure to DIY robots makes these robots feel safer. This finding is particularly relevant, as children's feelings of safety in HRI are rarely considered, but ought to play an important role in the responsible deployment of such technology. As both changes in perceptions occur during the first four weeks of the interaction but stabilize after, our findings suggest that these first encounters are especially critical for shifts in perceptions. An open question, however, remains about how many interactions are needed for these changes to occur. Previous work indicates that perception may fluctuate over repeated sessions with a robot and that different perceptual dimensions stabilize within different time frames (Paetzel et al., 2020). Future work should assess how students' perceptions of robots' socialness change across multiple interactions and the time frame over which perceptions may stabilize.

4.4 Anthropomorphism (and attachment) as a predictor of socialness perception, attachment, and intention to use

Our exploratory analyses suggest that perceived anthropomorphism is a central pillar of students' experience with a social robot, predicting their perceptions of competence, warmth, discomfort, safety, attachment, and intention to use. Additionally, we found that attachment was positively related to students' intention to use the robot. These findings are consistent with prior work showing that anthropomorphism mediates favourable judgements of robots across age groups, including prosocial behaviours (4-9 years, Nijssen et al., 2021), socialness (7-14 years, Barco et al., 2020; 19-26 years, Jastrzab et al., 2024; 8-14 years, Tung, 2016), appearance (8-14 years, Tung, 2016), and attachment (8-10 years, van Straten et al., 2023). Related work on chatbots also found that adults' attachment increases their intention to use them (Zhang & Yu, 2025). Our work adds to this body of work by highlighting that, even for older children and adolescents, anthropomorphism remains a robust driver of perceptions of socialness and that attachment predicts long-term engagement. However, our findings also suggest that anthropomorphism increases perceived discomfort. This finding may suggest that uncanny valley effects are present even with low-level anthropomorphic robots during extended interactions. Further research is needed to explore the uncanny valley effect in DIY robotic toolkits. We additionally propose that future research on students' interactions with robots should account for their perceived anthropomorphism of (and attachment to) specific robots when assessing other socialness perceptions.

4.5 Limitations

Conducting long-term, ‘in-the-wild’ HRI studies will always present a range of expected and unexpected challenges (Woo et al., 2021), which must be considered when interpreting findings. For this particular study, we encountered technical difficulties that introduced unintended experiential variance between the control and experimental groups. Specifically, connectivity issues with the robot’s LLM occurred within the control group sessions, resulting in these participants spending no time in verbal interactions with Marty. Instead, we shifted their focus towards more Arduino programming and alternative LLM activities (i.e., chatting with an alternative LLM not connected to Marty). This significant difference between groups may have contributed to the null findings. However, these technical errors also realistically demonstrate how robotic platforms are likely to fail in real-world deployments, an issue consistently reported in other studies (e.g., Woo et al., 2021). Furthermore, the pace of the curriculum varied across classrooms, resulting in some classes not completing all intended content within a given session (e.g., in session 5, the experimental group did not compare multiple prompts and responses from an LLM, whereas the control group did). These deviations, while reflective of real-world classroom settings, introduce noise into our longitudinal data.

Additionally, the physical room setup necessitated that teachers teach the experimental group sessions (Nov – Dec 2025) before the control group sessions (Jan – March 2026). This sequence may have introduced order effects, which we cannot entirely rule out. We also acknowledge the challenges of data quality in school-based research. Although we included attention checks and emphasized the importance of careful completion, we observed that survey completion times varied between pre-, mid-, and post-test. While we considered linking assessments to in-class grades to encourage more thoughtful completion, we ultimately decided against it to avoid placing undue stress on students. However, future research may consider this as a protection against careless survey completion.

Finally, our results are constrained by the specific humanoid morphology of the Marty robot, the context and tasks during our curriculum, the relatively small sample size, and the demographic and socioeconomic background of this sample. Our sample was drawn from a single German school, and participants’ gender was significantly unbalanced, with more boys participating than girls. These factors limit the generalizability of our findings to other robotic platforms or more diverse cultural and educational contexts. We are currently extending this line of research with a different robot in a different demographic context, and encourage others to join these efforts to replicate and broaden our findings.

4.6 Conclusion

Our work joins that of Marsh and colleagues (2018) in challenging the assumed universality of the IKEA effect, finding no evidence of it in older children’s and adolescents’ interactions with social robots. Additionally, we provide some of the first evidence of how anthropomorphism and safety perceptions develop over time in long-term interactions with teenage students. Specifically, our findings

confirm the moderating role of anthropomorphism in long-term educational interactions and suggest that initial exposure to robots can increase students' sense of safety with the robot. Lastly, our findings suggest that the positive side effects of DIY assembly (i.e., lower costs, increased accessibility, and support for students' digital literacy) do not come at the expense of the robot's social effectiveness. Our results thus support a recommendation to further explore DIY social robots in classrooms, as they may offer a viable and scalable alternative to traditional, expensive robotic platforms.

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